

Robust Gait Phenotyping Across Pathologies (IMU)

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Abstract—Wearable inertial measurement units (IMUs) enable scalable, real-world gait analysis for diverse clinical populations. In this study, we analyze the multi-pathology dataset by Voisard et al. to characterize gait patterns across neurological and orthopaedic conditions and to evaluate optimal sensor configurations. A pipeline was developed to extract clinically relevant temporal gait parameters, a standardized extraction of seven temporal and biomechanical gait parameters was performed, and a sensor ablation study was conducted using SVM, XGBoost, and Random Forest models. Results show that neurological disorders exhibit increased stride variability, reduced cadence, and decreased symmetry compared to healthy and orthopaedic groups. Across all models, the right foot (RF) sensor consistently provided the highest single-sensor performance, achieving up to 88.9% balanced accuracy. Two-sensor configurations offered the best trade-off between accuracy and cost, while additional sensors did not consistently improve performance. These findings highlight the potential of IMU-based gait metrics as robust, interpretable biomarkers and support the development of cost-effective, clinically deployable gait monitoring systems.

I. INTRODUCTION

Human walking is among the most complex and precisely regulated motor behaviors executed by the central nervous system. What appears as an effortless, automatic rhythm is in fact the emergent product of a multilevel neuromotor architecture: cortical motor planning, cerebellar timing, basal ganglia rhythmogenesis, spinal central pattern generators, and continuous proprioceptive and vestibular feedback [1]. The fidelity of this architecture is reflected in the precise bilateral coordination of left and right limb cycles, and its degradation is a sensitive early indicator of diverse neurological and orthopaedic conditions [1, 2].

The increase in aging population has increased the number of gait-associated morbidities related to orthopedic and neurological disorders such as osteoarthritis and Parkinson's diseases [3]. These disorders impair mobility, erode autonomy, increase susceptibility to falls, and premature death [3]. As a result, proactive gait surveillance and early anomaly detection are significant strategies to prevent these disorders and also find treatments early [2,3].

Quantitative gait analysis has long been established as a clinical tool, but its reach has historically been limited by the cost and spatial constraints of laboratory-based force-platform and motion-capture systems. Inertial measurement units (IMUs) have disrupted this paradigm: lightweight, wireless, and costing a fraction of laboratory systems, they enable gait measurement in clinics, rehabilitation wards, and home environments. The dataset published by Voisard et al. in *Scientific Data* (2025) represents one of the most comprehensive publicly available multi-pathology IMU gait resources to date: 1,356 trials from 260 participants across

three clinical categories (healthy, neurological, orthopaedic) and seven specific diagnostic subtypes, recorded with four synchronised IMUs and accompanied by structured gait event annotations and standardised clinical scoring. This richness of clinical metadata pathology-specific severity scores, physician-rated Visual Gait Assessment, and Timed Up and Go measurements, provides an exceptional foundation for validating computational gait biomarkers against established clinical standards. This paper reports the statistical validation of that hypothesis across all seven diagnostic subtypes, its clinical screening performance, and its relationship to concurrent machine learning classification pipelines.

II. RESEARCH QUESTIONS

1. RQ1 (Sensor Optimization): Which sensor configurations provide optimal cost-performance trade-offs for pathology classification across gait phases?
2. RQ2 (Temporal Asymmetry Hypothesis): Does temporal gait asymmetry differ between healthy and pathological populations in a direction opposite to classical assumptions?

III. MAGNITUDE OF THE PROBLEM

Gait disorders affect hundreds of millions of people worldwide and represent one of the most disabling clinical presentations in both neurology and orthopaedic surgery. Neurological conditions alone affect more than one billion people globally, with motor and coordination deficits that directly impair walking function across the lifespan [4].

Parkinson's disease (PD)-one of the five pathological cohorts in our dataset, affects approximately 8.5 million people worldwide, with prevalence projected to double by 2040 as the global population ages [5]. Characteristic gait changes including shuffling, reduced stride length, and freezing of gait are present in the majority of patients by mid-stage disease, contributing directly to disability and fall risk [3, 5]. Cerebrovascular accident (CVA/stroke) affects approximately 13 million people annually: among the 80 million stroke survivors worldwide, 30–40% exhibit persistent gait abnormalities severe enough to restrict community ambulation [6]. Radiation-induced leukoencephalopathy (RIL) and chemotherapy-induced peripheral neuropathy (CIPN), the remaining two neurological cohorts in the dataset, represent growing populations as cancer survival rates improve, with CIPN developing in 30-40% of patients receiving neurotoxic regimens [7].

On the orthopaedic side, knee osteoarthritis (KOA) affects an estimated 250 million people globally and is the leading cause of pain-related mobility impairment in adults over 50 [8]. Hip osteoarthritis (HOA) affects a further 15% of adults over 60 [3, 8]. Anterior cruciate ligament (ACL) injuries number approximately 200,000 annually in the United States

alone [9]. All three conditions alter gait mechanics in measurable but distinct ways, providing a complementary contrast to the neurological cohorts in our dataset.

The most serious and costly downstream consequence of gait impairment is falling. Falls are the leading cause of injury-related mortality among adults aged 65 and older [3, 5]. In the United States, the CDC estimates approximately 36 million falls annually among older adults, resulting in over 32,000 deaths, more than 3 million emergency department visits, and total healthcare costs exceeding \$50 billion per year [10]. PD patients fall at approximately three times the rate of age-matched healthy controls; post-stroke patients who fail to restore symmetric gait patterns have significantly longer rehabilitation stays and worse functional outcomes [5, 6].

IV. RELEVANCE

Wearable IMU-based gait analysis directly addresses three key unmet clinical needs: screening, monitoring, and rehabilitation assessment. For screening, a validated single-feature gait metric derived from a 10-metre walk could identify patients requiring specialist referral in primary care, community pharmacy, or remote telehealth settings—particularly in lower-income regions where access to specialists is limited. For longitudinal monitoring, objective IMU-derived metrics provide sensitive and reproducible endpoints for clinical trials of disease-modifying therapies, offering advantages over subjective rating scales that are susceptible to placebo effects and inter-rater variability [11].

The dataset by Voisard et al. (2025) was specifically designed to support this translational objective. Its standardized 10-metre walk protocol, four-sensor configuration, bilateral gait event annotations, and inclusion of pathology-specific clinical severity scores (UPDRS-III, mRS, FAC, TNS, Kellgren–Lawrence, VGA) enable direct validation of computational gait features against established clinical measures [3, 11]. Such integration of high-quality sensor data with detailed clinical metadata is uncommon in publicly available gait datasets, making this cohort particularly valuable for the development and validation of digital biomarkers.

V. WHAT HAS BEEN DONE WITH THE DATASET

a. *The Voisard et al. (2025) dataset*

The dataset was assembled at two French military hospital sites by a multidisciplinary team from Centre Borelli (Université Paris Saclay / Université Paris Cité) and the Service de Santé des Armées. It contains 1,356 trials from 260 subjects: 73 healthy (HS; ages 18–87), 44 orthopaedic (HOA, KOA, ACL), and 143 neurological (CVA, PD, CIPN, RIL). Four synchronised IMUs (100 Hz) were placed at the head (HE), lower back L4/L5 (LB), and bilateral feet (LF, RF). The dataset provides structured bilateral heel-strike and toe-off annotations and pathology-specific clinical severity scores, making it one of the richest publicly available multi-pathology IMU gait resources.

The dataset's creators validated the automatic gait event detection algorithm underpinning the heel-strike annotations in a dedicated methodological paper [12]. This work demonstrated high detection precision from healthy to

severely impaired patients, including festinating PD and spastic CVA gait, the methodological foundation on which our parameter extraction rests. The same group also developed the Semiogram-a multidimensional gait visualisation tool aggregating IMU-derived parameters across five clinical domains (speed, regularity, symmetry, stability, smoothness), which was validated in a multiple sclerosis cohort using the same acquisition protocol [13].

b. *Al-Harathi (2026) -G-MASA-TCN deep learning framework*

The first and most substantial downstream study to use this dataset is by Al-Harathi et al. (2026), which applied a deep learning Gait Multi-Anchor Space-Aware Temporal Convolutional Network (G-MASA-TCN)- integrating multi-scale temporal fusion, graph-informed spatial modelling, and residual dilated convolutions to the full 1,356-trial dataset across all eight cohorts. Trial lengths were standardised by zero-padding and truncation given the wide duration range (2,490 to 18,639 frames). Integrated Gradients explainability analysis identified foot-mounted IMU channels as the most attribution-weighted inputs, converging with our sensor ablation findings that RF is the single most informative sensor placement [3].

c. *Sadeghsalehi (2025) - attention-based framework and dataset audit*

The second key study is Sadeghsalehi (2025), which applied a multi-stream attention model to four binary classification tasks: PD screening, OA screening, post-stroke asymmetry detection, and PD-vs-CVA differential diagnosis. A critical contribution of this work is its role as an automated dataset auditor: attention weights assigned to left and right foot sensors revealed a severe laterality confound- all 15 HOA patients with unilateral pathology were right-sided (right/left ratio: 15/0), and 47 of 49 CVA patients had right-dominant motor deficits. This confound means that right-foot sensor dominance in those binary tasks reflects dataset-specific laterality bias rather than a general superiority of RF sensing [11].

d. *Gait Asymmetry in the literature*

The assumption that healthy walking is perfectly symmetric was challenged by Sadeghi et al. [14], who demonstrated through meta-analysis that able-bodied adults exhibit reliable, consistent left-right differences in multiple gait parameters driven by motor lateralization. The dominant limb generates greater propulsive impulse and slightly longer stride duration, producing a positive signed asymmetry index. Herzog et al. similarly found unexpectedly large symmetrical index values in young healthy adults, concluding that healthy human gait is inherently asymmetrical [15]. A 2025 framework paper by Mancini et al. formalizes this, recommending swing or step time asymmetry as a minimum required gait domain for Parkinson's disease monitoring alongside pace, rhythm, variability, and turning [16].

Gait asymmetry differs markedly between healthy individuals and pathological populations, with the magnitude and underlying mechanisms varying by condition. In healthy gait, symmetry is largely preserved with only minor physiological asymmetries reflecting intact bilateral coordination [14]. In contrast, neurological disorders such as

Parkinson's disease (PD) and stroke exhibit significantly greater asymmetry due to central nervous system dysfunction. PD is characterized by reduced stride length, increased variability, and impaired bilateral coordination, resulting in moderate-to-high asymmetry that worsens with disease progression [17, 18]. Stroke produces the most pronounced asymmetry, driven by hemiparesis, with substantial differences between paretic and non-paretic limbs in step length, stance time, and swing time, often persisting despite rehabilitation [19]. In contrast, orthopaedic conditions such as knee and hip osteoarthritis (KOA, HOA) show moderate asymmetry primarily due to pain-avoidance and mechanical compensation rather than disrupted motor control, with asymmetry often correlating with disease severity and joint loading patterns [20]. Notably, bilateral HOA may even present with relatively symmetric gait due to compensatory redistribution across both limbs. Peripheral conditions such as chemotherapy-induced peripheral neuropathy (CIPN) demonstrate a different pattern, where bilateral sensory impairment leads to reduced stride length and increased variability, often resulting in a more symmetric but less stable and less predictable gait pattern. Finally, anterior cruciate ligament (ACL) injury shows minimal asymmetry during steady-state walking, as the central gait program remains intact and deficits are more evident during high-demand tasks [21]. Overall, gait asymmetry is most severe in neurological conditions due to impaired central coordination, while orthopaedic and peripheral conditions primarily reflect compensatory adaptations with preserved motor control.

VI. METHODOLOGY

a. Data preparation and trial selection

We downloaded the full Voisard et al. (2025) dataset from Figshare (DOI: 10.6084/m9.figshare.28806086) and processed all 1,356 trials from 260 subjects. U-turn segments were excluded from all parameter computations, as they represent biomechanically distinct, non-steady-state gait:

Include event if: $HS < \text{urn_start}$ OR $HS > \text{urn_end}$

After exclusion, **974 valid straight-walking trials from 216 subjects** were retained. The cohort breakdown is shown in Figures 1 and 2.

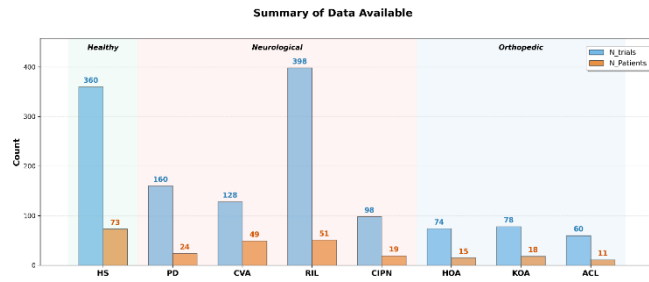


Fig. 1: Data availability by pathology group. Blue = number of gait trials; orange = unique subjects. All retained trials met good or acceptable quality criteria (≥ 3 valid gait cycles per limb). HS = Healthy; PD = Parkinson's Disease; CVA = Cerebrovascular Accident; RIL = Radiation-Induced Leukoencephalopathy; CIPN = Chemotherapy-Induced Peripheral Neuropathy; HOA = Hip OA; KOA = Knee OA; ACL = Anterior Cruciate Ligament Injury.

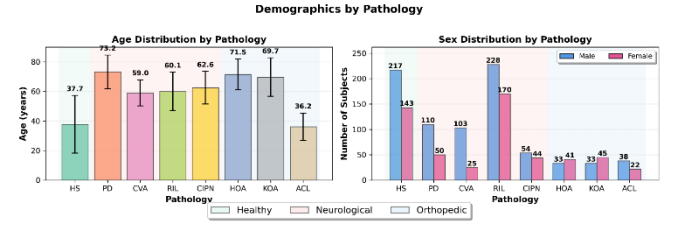


Fig. 2: Demographic characteristics. (A) Age distribution (mean $\pm$ SD) per group. (B) Sex distribution (blue = male, red = female). Background shading: green = Healthy, salmon = Neurological, sky blue = Orthopaedic.

b. Gait parameter extraction pipeline

We implemented a parameter extraction pipeline computing seven clinically validated gait metrics, including stride time, step time, cadence, stance phase, swing phase, stride CV, and symmetry index from the pre-annotated bilateral heel-strike timestamps. All parameters were computed per trial and aggregated at the subject level (mean across retained trials) to avoid pseudoreplication in group comparisons.

c. Sensor Ablation

A sensor ablation study was performed using the three ML models: XGBoost, SVM, Random Forest, by conducting comprehensive evaluation across different gait phases, window and overlap configurations, and sensor combinations. Features were extracted from each gait phase (full gait, pre-urn, post-urn, and u-turn) using different window sizes (1 - 6 s) and overlap (0%, 25%, and 50%). Performance was primarily assessed using balanced accuracy and macro-F1 score to account for class imbalance.

VII. PRELIMINARY RESULTS

a. Gait parameters across pathology groups

Figure 3 shows group-level gait parameters across all eight cohorts. Neurological groups, particularly PD, CVA, and RIL exhibit elevated stride times, decreased cadence, increased temporal variability (CV), and reduced symmetry compared to healthy subjects. Orthopaedic groups generally show intermediate or near-healthy values, with ACL patients indistinguishable from healthy controls on most temporal parameters, consistent with an intact central motor programme driving a mechanically altered but neurally intact gait.

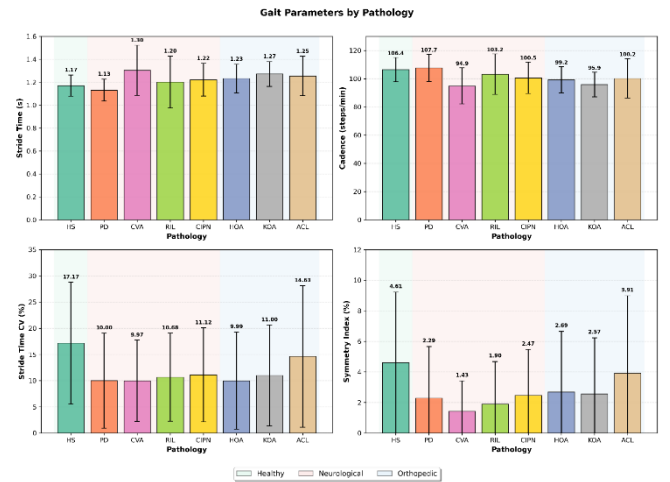


Fig. 3: Gait parameters across pathology groups (mean $\pm$ SD). (A) Stride time. (B) Cadence. (C) Stride time variability (CV). (D) Symmetry.

index. Green = Healthy; red = Neurological; blue = Orthopaedic.
Background shading delineates clinical categories.

The elevated CV in neurological groups (particularly PD and RIL) is consistent with the well-established literature linking stride time variability to neuromotor control integrity and fall risk. The reduced symmetry index in CVA patients reflects the expected hemiparetic gait pattern, though interpretation must account for the right-dominant laterality confound identified by Sadeghsalehi et al., 47 of 49 CVA patients in this dataset have right-hemisphere lesions, biasing the directionality of asymmetry measurements.

b. Temporal Asymmetry

(1) Healthy Vs Pathology

The central finding is that healthy subjects exhibit significantly larger temporal asymmetry than pathological groups (Table 1 and figure 4). Stride |AI| yields Cohen's $d = 0.77$ (95% Bootstrap CI: 0.50–1.07) with $p < 0.001$. AUC = 0.716 (95% CI: 0.635–0.792) at threshold 0.049, delivering 83% specificity and 59% sensitivity. Mean step time does not differ between groups ($p = 0.19$), ruling out speed as a confound.

TABLE I. SUBJECT-LEVEL TEMPORAL ASYMMETRY METRICS. 95% BOOTSTRAP CI SHOWN FOR PRIMARY STATISTIC. *** $p < 0.001$, ** $p < 0.01$ (WELCH'S T-TEST, HEALTHY VS. COMBINED PATHOLOGICAL).

Metric	Healthy (n=70)	Ortho (n=35)	Neuro (n=111)
Stride AI	0.052 ± 0.036	0.039 ± 0.027	0.029 ± 0.018
Stride L-R (s)	0.069 ± 0.054	0.054 ± 0.037	0.038 ± 0.023
Step AI	0.148 ± 0.091	0.088 ± 0.075	0.118 ± 0.093
Step CV (L)	0.284 ± 0.236	0.176 ± 0.153	0.187 ± 0.152
Mean step time	0.605 ± 0.035	0.628 ± 0.053	0.608 ± 0.057

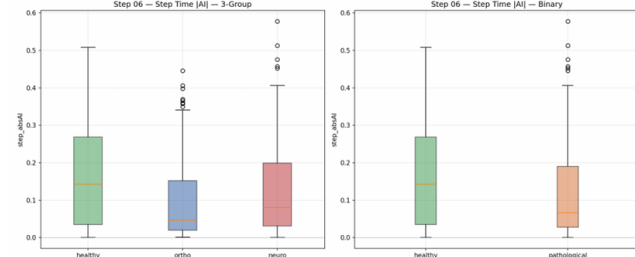


Fig. 4: Unsigned stride asymmetry index (|AI|) across three groups (left) and binary healthy vs pathological (right). Healthy subjects show significantly higher asymmetry magnitude ($d=0.77$).

In healthy individuals the signed AI is consistently positive (left stride slightly longer, mean = +0.009), reflecting dominant-leg lateralisation. In neurological patients it approaches zero and becomes directionally unpredictable. The orthopaedic group is not significantly different from healthy (LME $p = 1.00$), confirming mechanical joint pathology does not disrupt central lateralisation.

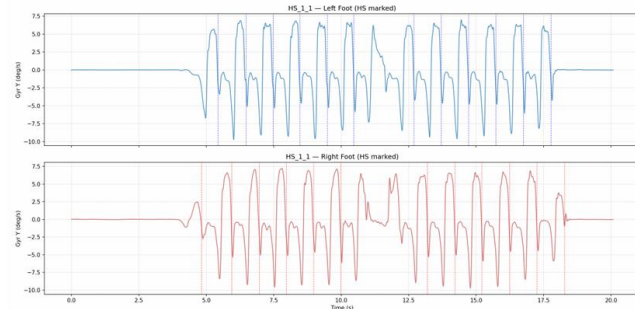


Fig. 5: Example trial showing left foot (blue) and right foot (orange) gyroscope Y-axis signals with heel-strike markers. The temporal structure of gait events is clearly visible

(2) Cross Pathology

Neurological conditions with central motor involvement show the largest asymmetry reductions (RIL $d=0.87$, PD $d=0.77$, CVA $d=0.73$, all $p < 0.001$). CIPN shows intermediate effects ($d=0.53$). KOA reaches significance ($d=0.45$, $p=0.034$). HOA and ACL are indistinguishable from healthy. The Sadeghsalehi et al. laterality confound applies to CVA and HOA (Table 2).

TABLE II. STRIDE |AI| BY SUBTYPE. † CVA AND HOA SUBJECT TO RIGHT-DOMINANT LATERALITY SAMPLING CONFOUND; INTERPRET WITH CAUTION.

Subtype	Category	Stride AI	d vs Healthy	p-value
RIL (n=14)	Neurological	0.024	0.87	<0.001***
PD (n=17)	Neurological	0.025	0.77	<0.001***
CVA (n=44)†	Neurological	0.027	0.73	<0.001***
CIPN (n=36)	Neurological	0.033	0.53	0.003**
KOA (n=14)	Orthopaedic	0.036	0.45	0.034*
HOA (n=12)†	Orthopaedic	0.042	0.27	0.226 ns
ACL (n=9)	Orthopaedic	0.049	0.09	0.779 ns

(3) ROC Screening performance and clinical validation

Stride |AI| achieves AUC = 0.716 [0.635–0.792] at threshold 0.049, sensitivity 0.59, specificity 0.83. Spearman correlation between VGA severity and stride |AI| is $\rho = -0.206$ ($p < 0.001$, $n = 779$), confirming *convergent validity* with established clinical assessment (Figures 6 and 7).

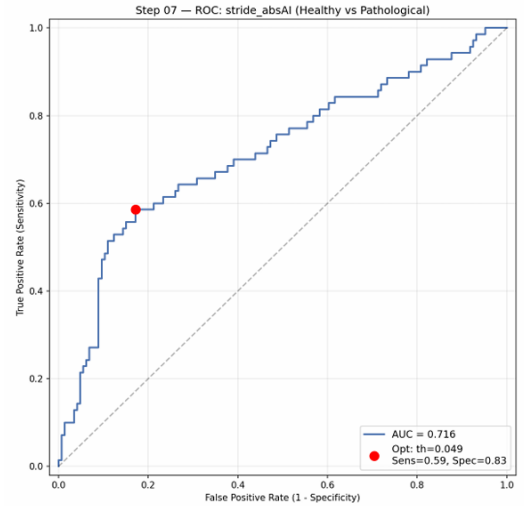


Fig 6: ROC curve for stride |AI| as a binary classifier (healthy vs pathological). AUC = 0.716 with an optimal Youden's J threshold at 0.049

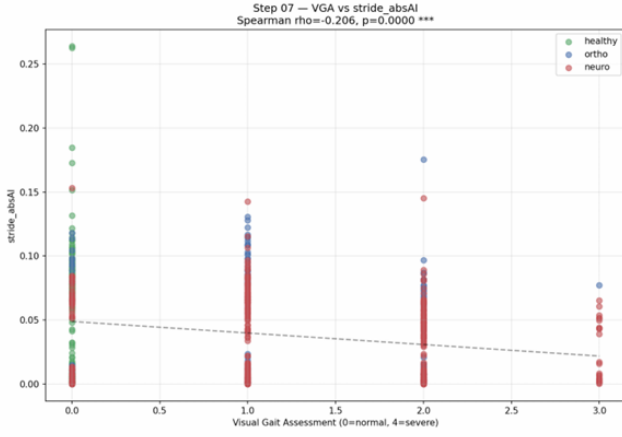


Fig 7: Scatter plot of Visual Gait Assessment score vs stride |AI|, colored by group. The negative correlation confirms that more severe gait impairment is associated with reduced temporal asymmetry.

c. ML feature Integration

Adding 11 asymmetry features to the 217-feature sensor set produced marginal but consistent SVM improvement (Table 4). On the matched 974-trial subset, macro-F1 improved +0.007 and BAcc +0.009. Asymmetry features alone achieve ~50% F1 (vs. 33% chance). No asymmetry feature appears in the RF top-25, confirming spectral features already implicitly encode temporal lateralisation.

TABLE III. 5-FOLD STRATIFIEDGROUPKFOLD CV (SUBJECT-GROUPED). *VALUES AFTER SLASH = MATCHED 974-TRIAL SUBSET.

Configuration	Features	LR F1	RF F1	SVM F1
Sensor only	217	0.748	0.810	0.816 / 0.806*
Sensor + Asymmetry	228	0.741	0.810	0.822 / 0.812*
Feet + Asymmetry	120	0.735	0.770	0.791
Asymmetry only	11	0.447	0.466	0.491

d. Feature directionality

Of 182 features significant at $p < 0.05$ (Kruskal-Wallis), the dominant pattern is monotonic decrease $\downarrow H > O > N$ (67 features); spectral centroid frequencies and signal dynamics decrease systematically from healthy to neurological, reflecting slower and less variable gait. All top-20 features show 93–100% cross-subject concordance, confirming effects are present in nearly every individual subject rather than being driven by group averaging.

TABLE IV. TOP FEATURES BY KRUSKAL-WALLIS SIGNIFICANCE WITH DIRECTIONAL PATTERN AND H-VS-N CONCORDANCE (FRACTION OF NEUROLOGICAL SUBJECTS ON EXPECTED SIDE OF HEALTHY MEDIAN)

Feature	Direction	H-vs-N Concordance
duration_s	$\uparrow H < O < N$	100%
LB_Acc_Y_spec_centroid_hz	$\downarrow H > O > N$	97%
HE_FreeAcc_Y_spec_centroid_hz	$\wedge$ peak-ortho	97%
HE_FreeAcc_X_spec_centroid_hz	$\vee$ valley-ortho	98%
LB_Gyr_Y_spec_power	$\uparrow H < O < N$	97%
RF_FreeAcc_Z_spec_centroid_hz	$\wedge$ peak-ortho	95%
RF_Acc_X_std	$\downarrow H > O > N$	94%

e. Sensor Ablation

(1) SVM

Sensor placement significance was investigated across all gait phases, window sizes and overlap percentages. As shown in Table 5, the RF sensor provided the most informative data which achieved highest performance across all gait phases and segmentation configurations. The highest balanced accuracy and macro-F1 score achieved by the SVM model using a single sensor were 87.6% and 80.8%, respectively. These results were obtained using the RF sensor with a 6 s window and a 25% overlap over the post-uturn data. While studying single sensor data over the entire gait data achieved 85.97% best balanced accuracy and 85.4% macro-F1 score using the RF sensor data with a 5 s window size and 50% overlap. Studying the pre-uturn gait data resulted in highest balanced accuracy and macro-F1 score of 83.2% and 80.8%, respectively, using the RF single sensor data with 4 s and 25% overlap.

TABLE V. BEST SENSOR CONFIGURATIONS ACROSS DIFFERENT GAIT PHASES AND DATA SEGMENTATION USING SVM

Gait Phase	Best Sensor configuration	Window size	Overlap	Balanced accuracy	Macro-F1
Full gait	Best 1 sensor (RF)	5000ms	50%	85.97%	85.4 %
Full gait	Best 2 sensors (RF+HE)	5000ms	50%	83.6%	81.5%
Full gait	Best 3 sensors (LB+RF+HE)	5000ms	50%	84.8%	82.7%
Pre-uturn	Best 1 sensor (RF)	4000ms	25%	83.2%	80.8%
Pre-uturn	Best 2 sensors (LB+RF)	3000ms	25%	83.1%	76.6%
Pre-uturn	Best 3 sensors (LB+LF+RF)	6000ms	50%	83.7%	75.2%
Post-uturn	Best 1 sensor (RF)	6000ms	25%	87.6%	80.8%
Post-uturn	Best 2 sensors (LB+HE)	6000ms	50%	89.8%	84.4%
Post-uturn	Best 3 sensors (LB+RF+HE)	6000ms	25 %	88.5%	82.9%
U-turn	Best 1 sensor (RF)	1280ms	50 %	76.0%	75.3%
U-turn	Best 2 sensors (LB+RF)	1280ms	50 %	77.7%	74.8%
U-turn	Best 3 sensors (LB+LF+RF)	1280ms	50 %	77.3%	75.4%

(2) XGBoost

XGBoost results for sensor ablation study are summarized in Table 6. The RF sensor data provided highest accuracy while studying full gait (5 s and 50 % overlap) and u-turn phases (1 s and 50 % overlap) with 88.7% and 78.2% balanced accuracies , respectively. However, the highest overall accuracy using a single sensor was achieved using the LB sensor data in post-uturn phase with 3 s window size and 50% overlap, which resulted in 90.7% balanced accuracy and 79.4% macro-F1 score.

TABLE VI. BEST SENSOR CONFIGURATIONS ACROSS DIFFERENT GAIT PHASES AND DATA SEGMENTATION USING XGBOOST

Gait Phase	Best Sensor configuration	Window size	Overlap	Balanced accuracy	Macro-F1
Full gait	Best 1 sensor (RF)	5s	50%	88.7%	82.2%
Full gait	Best 2 sensors (LB+RF)	5s	50%	91.8%	85.1%
Full gait	Best 3 sensors (HE+LB+RF)	5s	50%	87.9%	80.6%
Pre-uturn	Best 1 sensor (HE)	6s	50%	90.0%	90.2%
Pre-uturn	Best 2 sensors (HE+LB)	6s	50%	90.0%	90.2%
Pre-uturn	Best 3 sensors (HE+LB+LF)	6s	50%	90.0%	90.2%
Post-uturn	Best 1 sensor (LB)	3s	25%	90.7%	79.4%
Post-uturn	Best 2 sensors (HE+LB)	6s	25%	93.9%	89.8%
Post-uturn	Best 3 sensors (HE+LB+LF)	6s	25%	92.9%	88.4%
U-turn	Best 1 sensor (RF)	1s	50%	78.2%	65.3%
U-turn	Best 2 sensors (HE+RF)	1s	50%	80.8%	67.7%
U-turn	Best 3 sensors (HE+LF+RF)	1s	50%	77.1%	64.7%

(3) Random Forest

Random Forest results indicate that the RF sensor provided the most consistently informative data, achieving the highest performance for a single sensor in three of the four gait phases (Table 7). The best single-sensor performance was 88.9% balanced accuracy and 85.6% macro-F1 score, obtained using the RF sensor with a 3 s window and 25% overlap on pre-uturn data. For the full gait analysis, the RF sensor achieved 85.1% balanced accuracy and 84.2% macro-F1 with a 5 s window and 50% overlap. Analyzing the post-uturn phase also showed strong results, with the RF sensor yielding 88.5% balanced accuracy and 82.0% macro-F1 using a 6000ms window and 50% overlap.

TABLE VII. BEST SENSOR CONFIGURATIONS ACROSS DIFFERENT GAIT PHASES AND DATA SEGMENTATION USING RF

Gait Phase	Best Sensor configuration	Window size	Overlap	Balanced accuracy	Macro-F1
Full gait	Best 1 sensor (RF)	5000ms	50%	85.1%	84.2%
Full gait	Best 2 sensors (LB+HE)	5000ms	50%	87.0%	84.1%
Full gait	Best 3 sensors (LB+RF+HE)	3000ms	25%	88.6%	85.9%
Pre-uturn	Best 1 sensor (RF)	3000ms	25%	88.9%	85.6%
Pre-uturn	Best 2 sensors (LB+RF)	5000ms	25%	87.6%	83.6%
Pre-uturn	Best 3 sensors (LF+RF+HE)	5000ms	50%	88.1%	83.7%
Post-uturn	Best 1 sensor (RF)	6000ms	50%	88.5%	82.0%
Post-uturn	Best 2 sensors (LF+RF)	6000ms	50%	89.1%	82.7%
Post-uturn	Best 3 sensors (LF+RF+HE)	6000ms	50%	89.8%	83.7%

U-turn	Best 1 sensor (LB)	3000ms	25%	50.0%	44.4%
U-turn	Best 2 sensors (LB+LF)	3000ms	25%	50.0%	44.4%
U-turn	Best 3 sensors (LB+LF+RF)	3000ms	25%	50.0%	44.4%

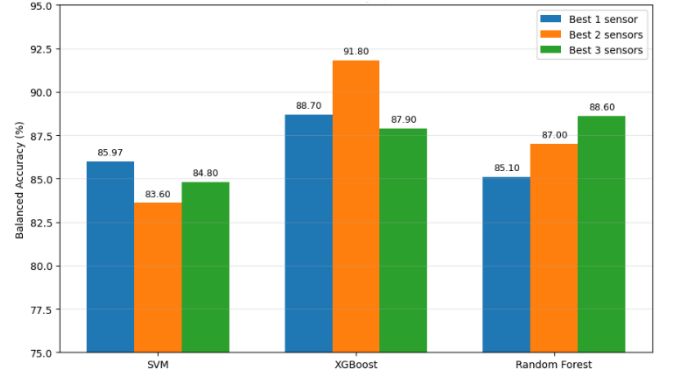


Fig. 8: Best balanced accuracy by model and sensor count in full gait data

VIII. DISCUSSION

Contrary to the common assumption that pathological gait is more asymmetric, we demonstrate that healthy subjects exhibit significantly greater temporal asymmetry ($d=0.77$, $p<0.001$). This finding aligns with Sadeghi et al.'s meta-analysis showing reliable left-right differences in able-bodied adults driven by motor lateralization [14]. Our data suggest that neurological conditions disrupt the lateralized motor program, causing convergence toward zero asymmetry. This reframing is clinically important: reduced asymmetry may be an early biomarker of neuromotor control degradation, potentially detectable before clinical symptoms emerge.

Clinical Utility: The $AUC=0.716$ at threshold 0.049 provides 83% specificity for ruling out pathology in primary care screening (Figures 6 and 7). While sensitivity is modest (59%), the high specificity means negative screens reliably identify healthy individuals, reducing unnecessary specialist referrals.

Across the analyses, the RF sensor is consistently the top recommendation for a single-sensor placement. Where the Random Forest, XGBoost, and SVM models achieve up to 88.9%, 88.7%, and 87.6% balanced accuracies, respectively (Figure 8), with the RF sensor alone, making it a robust choice for cost-effective applications. For a two-sensor setup, which offers an optimal balance of performance and cost, the XGBoost analysis identifies LB + RF as the best combination for full gait analysis, yielding 91.8% accuracy, while HE + LB combination achieves higher balanced accuracy of 93.9% in post-uturn analysis. For Random Forest models, effective pairs include LB + HE for full gait with 87.0% balanced accuracy and LF + RF for post-uturn with 89.1% accuracy. Results show that adding more than two sensors does not necessarily guarantee improved classification accuracy as

shown in Figure 4, that provides comparison of performance of the three models using the full gait data.

Window size and overlap significantly impact performance. Larger windows of 5-6 seconds are generally optimal for continuous gait phases across models. While the u-turn gait phase requires smaller window size to capture important features, however, studying the u-turn event alone is not recommended as it achieves the lowest performance across all gait phases. A 50% overlap is broadly recommended, as it improves classification stability and appears in high-performing RF configurations. However, optimal settings remain dependent on the specific gait phase being analyzed.

Five key findings: (1) Healthy gait is asymmetric (2)RF is the most informative single placement; (3) two sensors (HE+LB or RF+LB) capture near-ceiling performance; (4) three sensors show diminishing returns; (5) post-U-turn phase is consistently the most discriminative; (6) U-turn analysis should not be used as a standalone input.

IX. FUTURE WORK

A key step in our project is to extend the current three-class classification framework (healthy, neurological, orthopaedic) to full eight-class classification across all diagnostic subtypes: healthy, PD, CVA, RIL, CIPN, HOA, KOA, and ACL. This sub-class classification would provide clinically actionable subtype-specific predictions rather than broad category labels, enabling differential diagnosis support directly from IMU gait data. The expanded model will be evaluated using the same sensor ablation framework established in this work, allowing direct comparison of subtype-level performance against the current three-class results.

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